THE COMPLETION FUNCTOR IS AN EQUIVALENCE ON RIG-ETALE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7042-7097, following the statement at 7014-7040.

Editorial note. The final 'reader can check ... some details omitted' step is the sheaf-level gluing of already-established descent data, flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 7096 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Fully faithfulness of the completion functor. lemma-completion-functor-fully-faithful (carded L26155, cited at ~7043). Twice: for the a_ij and their cocycle condition.
- Affine dilatations. lemma-dilatations-affine (carded L26272, cited at ~7055). Supplies X'_i -> X_i with (X'_i)_{/T} = W_i.
- Effectiveness of descent data. Descent on Spaces, Definition spaces-descent-definition-descent-datum-for-family-of-morphisms and Bootstrap, Lemma bootstrap-lemma-descend-algebraic-space (cited at ~7072-7076). Descends the X'_i to X'.

Source reference. Original reference 0ADV: focused-seeds/geometry/S9-restricted-dilatations-general/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0AR5: sources/stacks-restricted/pinned-original-source.tex, source lines 6766–6766. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GDR: sources/stacks-restricted/pinned-original-source.tex, source lines 6832–6832. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ARA: sources/stacks-restricted/pinned-original-source.tex, source lines 6946–6946. Navigation pointer only: the counted proof cites

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this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GDQ: sources/stacks-restricted/pinned-original-source.tex, source lines 6790–6790. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ADI: focused-seeds/geometry/S9-restricted-dilatations-general/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1. *Let S be a scheme. Let X be a locally Noetherian algebraic space over S. Let $T \subset |X|$ be a closed subset. Let $U \subset X$ be the open subspace with $|U| = |X| \setminus T$. The completion functor (0AR5)*

$$\left\{\begin{array}{c}\text{morphisms of algebraic spaces}\\ f : X' \to X \text{ which are locally}\\ \text{of finite type and such that}\\ f^{-1}U \to U \text{ is an isomorphism}\end{array}\right\} \longrightarrow \left\{\begin{array}{c}\text{morphisms } g : W \to X_{/T}\\ \text{of formal algebraic spaces}\\ \text{with } W \text{ locally Noetherian}\\ \text{and } g \text{ rig-étale}\end{array}\right\}$$

sending $f : X' \to X$ to $f_{/T} : X'_{/T'} \to X_{/T}$ is an equivalence.

Proof. The functor is fully faithful by Lemma 0GDR. Let $g : W \to X_{/T}$ be a morphism of formal algebraic spaces with W locally Noetherian and g rig-étale. We will prove W is in the essential image to finish the proof.

Choose an étale covering $\{X_i \to X\}$ with X_i affine for all i. Denote $U_i \subset X_i$ the inverse image of U and denote $T_i \subset X_i$ the inverse image of T. Recall that $(X_i)_{/T_i} = (X_i)_{/T} = (X_i \times_X X)_{/T}$ and $W_i = X_i \times_X W = (X_i)_{/T} \times_{X_{/T}} W$, see Lemma 0GDQ. Observe that we obtain isomorphisms

$$\alpha_{ij} : W_i \times_{X_{/T}} (X_j)_{/T} \longrightarrow (X_i)_{/T} \times_{X_{/T}} W_j$$

satisfying a suitable cocycle condition. By Lemma 0ARA applied to $X_i, T_i, U_i, W_i \to (X_i)_{/T}$ there exists a morphism $X'_i \to X_i$ of algebraic spaces which is locally of finite type and an isomorphism over U_i and an isomorphism $\beta_i : (X'_i)_{/T} \cong W_i$ over $(X_i)_{/T}$. By fully faithfullness we find an isomorphism

$$a_{ij} : X'_i \times_X X_j \longrightarrow X_i \times_X X'_j$$

over $X_i \times_X X_j$ such that $\alpha_{ij} = \beta_j|_{X_i \times_X X_j} \circ (a_{ij})_{/T} \circ \beta_i^{-1}|_{X_i \times_X X_j}$. By fully faithfulness again (this time over $X_i \times_X X_j \times_X X_k$) we see that these morphisms a_{ij} satisfy the same cocycle condition as satisfied by the α_{ij}. In other words, we obtain a descent datum (as in Descent on Spaces, Definition 0ADI) (X'_i, a_{ij}) relative to the family $\{X_i \to X\}$. By Bootstrap, Lemma 0ADV, this descent datum is effective. Thus we find a morphism $f : X' \to X$ of algebraic spaces and isomorphisms $h_i : X' \times_X X_i \to X'_i$ over X_i such that $a_{ij} = h_j|_{X_i \times_X X_j} \circ h_i^{-1}|_{X_i \times_X X_j}$. The reader can check that the ensuing isomorphisms

$$(X' \times_X X_i)_{/T} \xrightarrow{\beta_i \circ (h_i)_{/T}} W_i$$

over X_i glue to an isomorphism $X'_{/T} \to W$ over $X_{/T}$; some details omitted. $\square$